



VANILLIN

The Chemistry Behind the Smell of Vanilla



1 IDENTITY

IUPAC Name:

4-hydroxy-3-methoxybenzaldehyde

Formula: $C_8H_8O_3$

Molar Mass: 152.15 g/mol

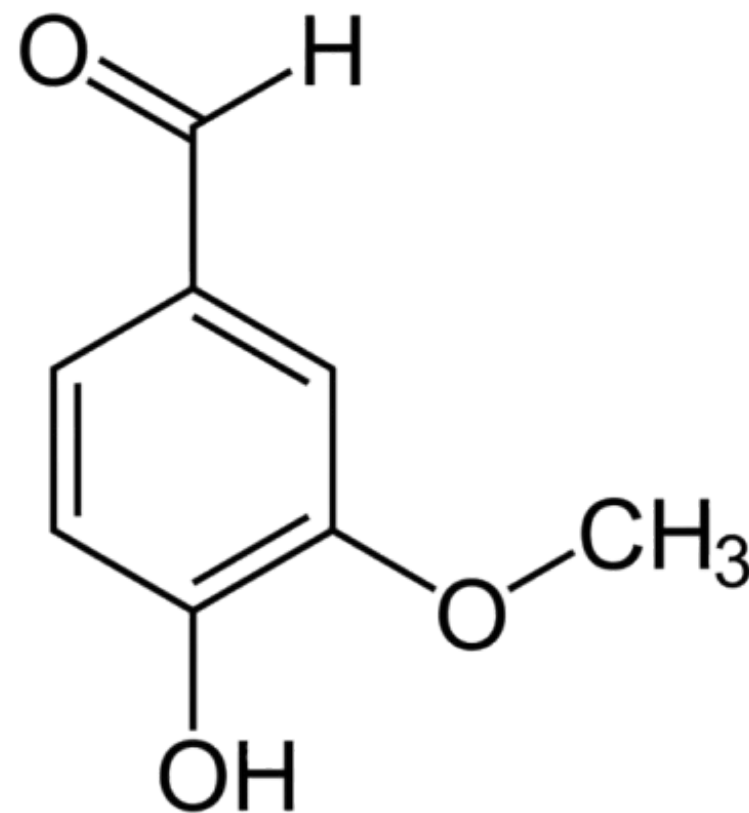
Found in:

Vanilla beans, flavors, and fragrances

2 STRUCTURE & IDENTITY

Functional Group:

- Aldehyde ($-CHO$)
- Hydroxyl ($-OH$)
- Methoxy ($-OCH_3$)
- Aromatic benzene ring



COOL CHEMISTRY FACT

The position of these groups on the ring creates the sweet, warm vanilla aroma. Change the structure, change the smell.

3 IMPORTANT PROPERTIES

White to pale yellow crystalline solid

Melting point: $\sim 81-83^\circ C$

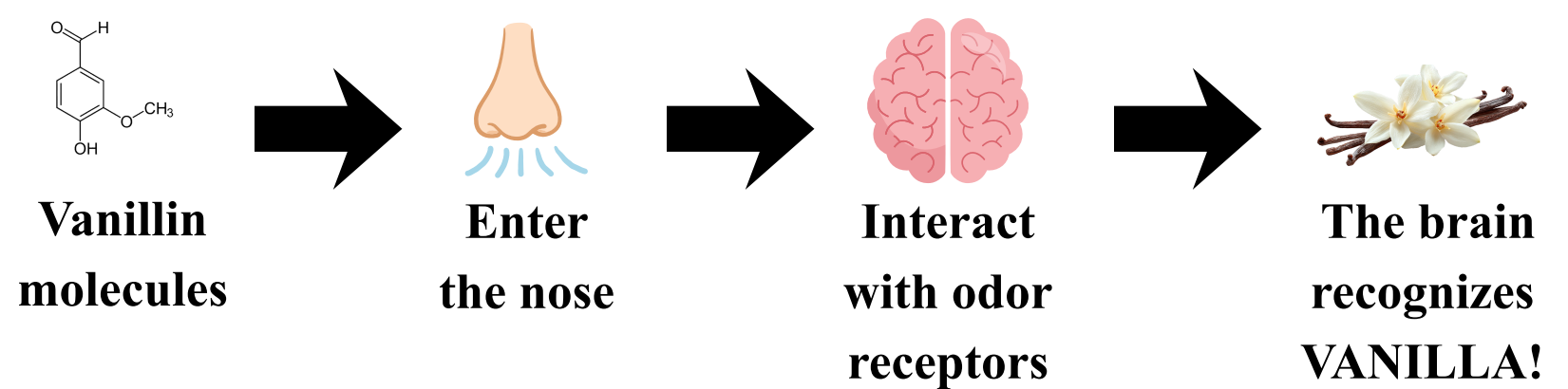
Sweet, warm vanilla odor

Slightly soluble in water

More soluble in organic solvents

4 WHAT MAKES IT AMAZING?

SMELL IS MOLECULAR RECOGNITION!



A small change in molecular structure can create a completely different smell

5 USES & VALUE

- Flavoring in foods (ice cream, chocolates, baked goods)
- Perfumes and cosmetics
- Soaps and personal care products

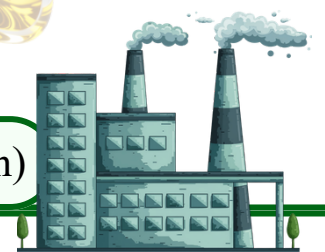
- Food and beverage industry
- Chemical and pharmaceutical applications



NATURAL VANILLIN (from plants)

OR

SYNTHETIC VANILLIN (industrial production)



6 RISK & SAFETY



- May cause skin or eye irritation
- Avoid inhalation of dust

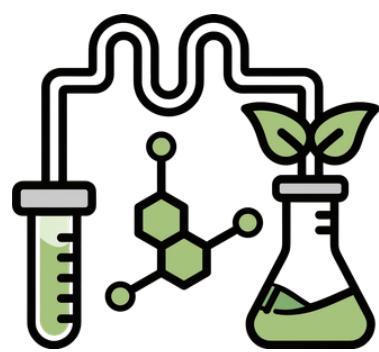
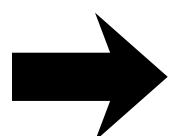
- Handle using appropriate protective equipment
- Work in a well-ventilated area

7 CHEMICAL ENGINEERING CONNECTION



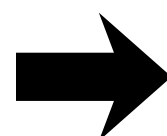
EXTRACTION

Obtain vanillin from natural materials



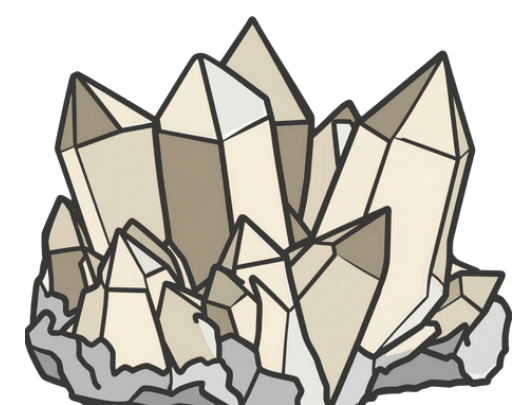
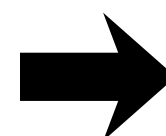
SYNTHESIS

Produce vanillin from chemical starting materials



SEPARATION

Remove impurities using distillation, extraction, etc.



CRYSTALLIZATION

Recover purified vanillin crystals

8 REFERENCE



- National Center for Biotechnology Information. (n.d.). PubChem compound summary for CID 1183, vanillin. PubChem.